single “electroweak” force. But particles were starting to emerge from the primordial soup—specifically, quarks and gluons.

Quarks, these days, are most commonly encountered as the components of protons and neutrons (which, together, are called hadrons). Gluons are the “glue” that binds quarks together via the strong nuclear force, and they are aptly named. They’re so good at binding quarks together that while quarks have been found in twos or threes or even occasionally quads and quintuples, finding a single quark in isolation has so far proved impossible. It turns out that if you have two quarks bound together (in an exotic particle called a meson), you have to put in so much energy to separate the quarks that before you can get them apart, the energy you just expended spontaneously produces two more quarks. Congrats! Now you have two mesons.

In the very early universe, however, the usual rules didn’t apply any more to singleton quarks than they did to anything else. Not only were the forces of nature operating under different laws, the universe contained a different mix of particles, and temperatures were so high that bound states of quarks could not exist in a stable form. Quarks and gluons bounced around freely in a hot roiling mix called a quark-gluon plasma—kind of analogous to the inside of a fire, but nuclear.

This “quark era” lasted until the universe reached the ripe old age of a microsecond. Meanwhile, somewhere in there (probably around the 0.1 nanosecond mark) the electroweak force split into electromagnetism and the weak nuclear force. Also around that time, something happened to create a distinction between matter and antimatter (matter’s annihilation-happy evil twin), allowing most of the universe’s antimatter to annihilate away.[fn15] Exactly how and why that happened is still a mystery, but as matter we can be glad it occurred, so we’re not constantly running into antimatter particles and vanishing in a puff of gamma rays.

In contrast to the GUT era, we actually know quite a lot about the quark era and the quark-gluon plasma. The theory is pretty well developed and less of a departure from standard particle physics than a GUT, and experiments confirm the predictions we make when we start from electroweak theories and extrapolate from there. But the real coup is that we can actually *re-create* quark-gluon plasma in the lab. Particle colliders like the Relativistic Heavy Ion Collider (RHIC) and the Large Hadron Collider (LHC) can, by smashing together gold or lead nuclei at extremely high